GRAVITYSORT · DESIGN DECISION RECORD

GravitySort concept selection

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/decisions/0004-concept-selection.md @ d6d285d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline): three concept options scored against GVS-URS-001; all within 10 %, so the choice goes to Amish without a pick (Proposed, awaiting Amish)

0004: Concept selection

- **Date:** 2026-10-25
- **Status:** proposed (Proposed, awaiting Amish; GVS-DEC-001 O1)

Context

GravitySort was designed before kit 1.8, as a fluidized centrifugal bowl and a small shaking table on one pedal drive, and has been at TRL 3 since 2026-09-25. Kit 1.8 asks every repo for at least two concept options with different working principles, scored against the user requirements (STANDARDS section 21), and kit 1.9 asks for every option to be modelled to the repo's detail level (Amish, 2026-10-24: "Detail every option"). In a repo designed before kit 1.8 the current design is option A, and it is not redesigned without Amish's go. The options, their STEM rows, first-order numbers and the full scoring are in GVS-OPT-001.

Options considered

Option	Working principle	Weighted score (of 230)	Main strength	Main weakness
A. Fluidized bowl and shaking table (current design)	Enhanced gravity: a fluidized bed at about 60 G holds fine gold; the table cleans the concentrate to about 100 g	165	Catches fine gold (UR-003) and makes a smeltable concentrate	Dearest ($594, USD 139 over the target), most complex, spinning bowl and open containment item
B. Diaphragm jig and shaking table	Pulsed water through a ragging bed stratifies by density; same table for clean-up	157	Slow, simple, safe mechanism; cheaper (about $400, estimate)	Loses much of the gold finer than 0.1 mm; more water; more concentrate to clean
C. Carpeted sluice and gold pans	Flowing film over expanded metal riffles and matting; hand panning	172	Cheapest (about $195, estimate), no power, no moving parts	Under 20 % recovery of gold finer than 100 um; tens of kilograms to pan by hand, the material miners amalgamate today

All three are within 10 % of the top score (A is 4.1 % and B 8.7 % below C).

Decision

Proposed, awaiting Amish. Under STANDARDS section 21 two options within 10 % go to Amish without a pick, so no option is recommended. Two observations for the decision:

1. The weighted table treats no requirement as a gate. C scores highest because cost, buildability and safety are counted twice (as user requirements and as the section 21 criteria), while it scores 1 of 5 on UR-003, the fine-gold need the project exists for. If Amish treats UR-003 as a gate (a score of 3 or more), only option A passes.
2. Under section 21 the current design, option A, stays in the repo (model, drawings, BOM and build plan) unless Amish decides otherwise; `concept_chosen` stays unset.

Consequences

- If A is confirmed: no change to the repo; the open items are the TRL 4 tests (PR-004, PR-005, PR-016) and the decisions in GVS-DEC-001.
- If B or C is chosen: the repo would be redesigned around that option at TRL 2 (its D3 model in `cad/src/gvs_option_b.py` or `gvs_option_c.py` is the starting point), with new calculations, drawings, BOM and build plan; the pitch would change for C (no fine-gold capture), which is Amish's decision in any case.
- Ideas worth grafting whatever is chosen: a short carpeted sluice from C under the bowl's tailings outlet as a no-power scavenger for coarse gold and for upsets; B's slow, low-energy drive is a reminder that the table clean-up costs only about 15 W.